

Lecture 05 Notes

Main Theme: Existence, Uniqueness, and Vector Spaces

[Likely exam topic] The lecture frames **existence and uniqueness of solutions** for linear systems as the main problem. The instructor uses that problem as the reason to introduce and use vector space ideas.

The guiding question is:

Given a linear system $Ax = b$:

- Does a solution x exist?
- If a solution exists, is it unique?
- If not unique, how many solutions or what kind of solution set should we expect?

The instructor emphasizes that the answer will come from vector spaces and from two different ways of looking at matrix-vector multiplication.

Review: Vector Spaces

[Exam note] A **vector space** was reviewed as a mathematical object with four components:

- A set of vectors.
- A set of scalars.
- Vector addition.
- Scalar multiplication.

These objects must satisfy a collection of vector space properties. The instructor did not relist all axioms here, but emphasized the most important conclusion for the current lecture:

Linear combinations of vectors in a vector space stay inside the vector set.

In other words, if vectors belong to the vector space, then any expression of the form

$$\alpha_1 v_1 + \alpha_2 v_2 + \cdots + \alpha_k v_k$$

also belongs to the same vector space.

This closure under linear combinations is the property that connects vector spaces to span, basis, column space, and the existence of solutions to $Ax = b$.

Review: Subspaces

[Exam note] A **subspace** is a subset of a vector space that is itself a vector space.

The lecture recalled geometric examples:

- In $\mathbb{R}^2$, subspaces include the origin, lines through the origin, and $\mathbb{R}^2$ itself.
- In $\mathbb{R}^3$, the variety of subspaces increases: examples include lines through the origin, planes through the origin, the origin, and the full space.

Important warning:

A set may be a subset of a vector space without being a subspace. To be a subspace, it must satisfy the vector space requirements, especially closure and inclusion of the zero vector.

Matrix Subspace Examples

[Exam note] The lecture reviewed examples in spaces of square matrices.

Symmetric Matrices

The set of symmetric matrices forms a subspace of the square matrix space.

The relevant idea is that linear combinations of symmetric matrices remain symmetric.

Zero-Trace Matrices

The set of square matrices with trace zero forms a subspace.

The trace condition is compatible with linear combinations:

$$\text{tr}(\alpha X + \beta Y) = \alpha \text{tr}(X) + \beta \text{tr}(Y).$$

So if both X and Y have zero trace, then any linear combination also has zero trace.

Matrices Orthogonal to a Fixed Matrix

The instructor mentioned matrices X that make a zero trace expression when multiplied by some A^T . This is pointing toward a matrix inner product:

$$\langle A, X \rangle = \text{tr}(A^T X).$$

The course had not yet formally defined matrix inner products at this point, but the instructor explained the interpretation:

If

$$\text{tr}(A^T X) = 0,$$

then X is orthogonal to A under that matrix inner product.

This gives a nontrivial example of a subspace: all matrices orthogonal to a fixed matrix A .

Orthogonal Matrices Are Not a Subspace

The set of real orthogonal matrices satisfies

$$XX^T = I$$

or, depending on convention, $X^T X = I$.

This set is a subset of the square matrix space, but it is **not** a subspace.

Main reason given:

The zero matrix is not included, because $0 \cdot 0^T = 0$, not I .

So the set fails a basic requirement for being a vector space/subspace.

Span

The instructor reviewed the **span** of a set.

Mechanistic definition:

Given a set of vectors A , take all possible linear combinations of those vectors. The set of all those combinations is the span of A .

$$\text{span}(A) = \{\alpha_1 a_1 + \cdots + \alpha_k a_k : a_i \in A, \alpha_i \text{ scalars}\}.$$

Conceptual definition:

$$\text{span}(A)$$

is the smallest vector space that contains A .

Equivalently, no proper smaller vector space can contain all of A while being strictly smaller than $\text{span}(A)$.

The instructor also phrased this as span taking a smaller set of vectors and generating a bigger set that is itself a vector space, because all possible linear combinations are included.

This idea is important because the column space of a matrix will later be defined as the span of its columns.

From Span to Basis and Dictionary

[Exam note] The instructor shifted from the span question to the opposite question.

Span starts with a set of vectors and asks:

What vector space is generated by this set?

The opposite question is:

Given a vector space, what set of vectors can span it?

This motivates **basis** and **dictionary**.

The instructor's motivation was representation: even if a vector space contains infinitely many vectors, one may want to represent or parameterize the whole space using only a small generating set.

The instructor used the example of a plane through the origin:

- The plane can be represented as all linear combinations of two non-collinear vectors p_1 and p_2 .
- The same plane can also be represented as all linear combinations of three vectors p_1, p_2, p_3 , if all three lie in the plane and span it.

The set $\{p_1, p_2\}$ is minimal in the sense that two non-collinear vectors are enough to generate the plane.

The set $\{p_1, p_2, p_3\}$ is not minimal if one vector can be written as a linear combination of the others. For example,

$$p_3 = \alpha_1 p_1 + \alpha_2 p_2.$$

Then p_3 is redundant.

Because of that redundancy, one could remove p_3 and keep p_1, p_2 , or possibly remove p_2 and keep p_1, p_3 , as long as the remaining vectors still span the same plane. The important point is that a redundant spanning set contains at least one vector that is not essential.

Instructor remark:

At this stage, the instructor called $\{p_1, p_2\}$ a basis only intuitively, before giving the official definition later.

Dictionary Versus Basis

A **basis** has no redundancy.

A **dictionary** may have redundancy. The instructor used the spanning set $\{p_1, p_2, p_3\}$ for a two-dimensional plane as a dictionary example.

The instructor emphasized that dictionaries are not bad. In some applications, dictionaries are preferred because they allow **sparse representations**.

Non-Uniqueness in a Dictionary

If a vector x lies in the plane spanned by $\{p_1, p_2, p_3\}$, then it can be represented as

$$x = \alpha_1 p_1 + \alpha_2 p_2 + \alpha_3 p_3.$$

Because the dictionary is redundant, the coefficients $\alpha_1, \alpha_2, \alpha_3$ are not unique. There may be infinitely many coefficient choices for the same vector.

This non-uniqueness gives a degree of freedom. One can choose coefficients so that many of them are zero.

This is called **sparsity**.

Sparse Representation Example

The instructor said the sparsity idea is not very clear in the simple case of three vectors in a two-dimensional plane, but it becomes useful in high-dimensional settings.

Example:

- Suppose the space is 100-dimensional.
- Choose a dictionary with 150 vectors.
- Because the dictionary has more vectors than a basis would need, representations are redundant.
- The dictionary can be chosen based on the type of data of interest.
- Then vectors from the data may be represented with coefficient vectors that are mostly zero.

Relationship:

Redundancy creates non-uniqueness, and non-uniqueness can be used to impose sparsity constraints.

The instructor said this topic would be discussed later and presented it here mainly to give the flavor of dictionary versus basis.

Linear Independence

The restriction that a basis has no redundancy leads to **linear independence**.

English definition:

A set A is linearly independent if no element of A can be written as a linear combination of the other elements of A .

Interpretation:

Each element has value because it cannot be produced from the others.

Equivalently, no vector in the set is redundant.

The set $\{p_1, p_2, p_3\}$ from the plane example is not linearly independent if any of the vectors can be written using the other two. For example:

- p_3 may be a linear combination of p_1 and p_2 .
- Or p_2 may be a linear combination of p_1 and p_3 .

Algebraic Definition of Linear Independence

The more standard definition is:

A set of vectors is linearly independent if the only way a finite linear combination can equal zero is for all coefficients to be zero.

For vectors $v_1, \dots, v_k$, this means:

$$\alpha_1 v_1 + \dots + \alpha_k v_k = 0 \iff \alpha_1 = \dots = \alpha_k = 0.$$

Equivalently, there should be no nonzero coefficient choice that produces the zero vector.

For an infinite set, the same idea is checked through finite selections of vectors from the set: any finite nontrivial zero combination reveals redundancy.

Proof Idea for Equivalence

If a nontrivial linear combination equals zero, then at least one coefficient is nonzero. One can move that vector term to the other side and solve for the corresponding vector as a linear combination of the others.

Example from the lecture:

Suppose

$$2p_1 + p_2 = 0.$$

Then

$$p_2 = -2p_1.$$

So p_2 can be produced from p_1 , meaning the set is redundant and not linearly independent.

This explains why the zero-combination definition captures the “no redundancy” idea.

Basis

After defining linear independence, the instructor gave the official basis definition.

A set B is a **basis** of a vector space V if:

1. B spans V :

$$\text{span}(B) = V.$$

2. B is linearly independent.

So a basis is a spanning set with no redundancy.

The basis concept restricts redundancy. A dictionary may span the same space but fail to be a basis if it contains redundant vectors.

Dimension

The **dimension** of V is the cardinality of a basis of V .

For finite-dimensional spaces, cardinality means the number of basis elements.

For infinite-dimensional spaces, cardinality involves a notion of infinity.

The lecture focused on finite-dimensional examples, but the instructor noted the broader cardinality language.

Standard Basis for $\mathbb{C}^n$

For n -dimensional complex vectors, the standard basis consists of vectors

$$e_1, e_2, \dots, e_n.$$

Each e_i is zero everywhere except at the i -th location, where it equals one.

Example:

$$e_1 = (1, 0, \dots, 0)^T, \quad e_2 = (0, 1, 0, \dots, 0)^T.$$

These basis vectors align with the coordinate axes and have unit length.

In $\mathbb{R}^3$, these are the usual e_1, e_2, e_3 vectors along the three coordinate axes.

Standard Basis and Kronecker Delta

The instructor connected standard basis vectors to delta functions used in electrical engineering.

The instructor first mentioned Dirac delta language, then corrected to **Kronecker delta functions** for the discrete finite-dimensional setting.

If the vector index is interpreted as a discrete time axis:

- One standard basis vector corresponds to a delta at sample time 0.
- Another corresponds to a delta shifted by one sample.
- Others correspond to further time-shifted Kronecker deltas.

This matters because standard basis vectors can be interpreted as time-localized impulses in signal processing.

Coordinates

Coordinates are basis-dependent.

If a vector x is written as a linear combination of basis vectors,

$$x = x_1 e_1 + x_2 e_2 + \dots + x_n e_n,$$

then $x_1, \dots, x_n$ are the coordinates of x with respect to that basis.

When the basis is the standard basis, these are called **standard coordinates**.

For another basis $b_1, \dots, b_n$, the same vector can be written as

$$x = \alpha_1 b_1 + \dots + \alpha_n b_n.$$

Then $\alpha_1, \dots, \alpha_n$ are the coordinates of x with respect to that chosen basis.

Important relationship:

The same vector can have different coordinate representations depending on the basis.

Complex Exponential Basis for $\mathbb{C}^n$

The instructor introduced another important basis for $\mathbb{C}^n$: the complex exponential vectors.

There are n such vectors, denoted f_k .

The entries are of the form

$$(f_k)_i = e^{j2\pi(k-1)(i-1)/n}.$$

The transcript discussion involved some index clarification:

- Vector indexing in linear algebra often starts at 1.
- Time indexing in signal processing often starts at 0.
- The expression uses $i - 1$ so that the first vector entry corresponds to time/sample index 0.

The instructor noted this as a source of confusion.

Real and Imaginary Parts of Complex Exponentials

By Euler's identity,

$$e^{j\theta} = \cos(\theta) + j \sin(\theta).$$

Therefore:

- The real part of a complex exponential vector behaves like a sampled cosine.
- The imaginary part behaves like a sampled sine.
- The parameter k determines the frequency.

The instructor noted that one cannot directly draw $e^{j\theta}$ as a real-valued curve because it is complex-valued, but one can draw its real and imaginary components.

For each k , the cosine and sine have a different frequency as functions of the vector index.

Complex Exponentials Form a Basis

[Exam note / homework emphasis] The instructor said that for different k values, the vectors f_k are independent and span the n -dimensional space.

Therefore, the complex exponential vectors form a basis of $\mathbb{C}^n$.

This means any vector in n -dimensional complex space can be written as a linear combination of those complex exponential vectors.

The instructor said this would be examined in more detail in homework.

Bases Are Not Unique

The class discussed that every finite-dimensional vector space has infinitely many possible bases.

The complex exponential basis is just one choice.

Important relationship:

A basis is not unique, but each basis gives a coordinate system. Choosing a basis is choosing how to represent vectors.

The instructor then asked why one would focus on the complex exponential basis in particular.

Why Complex Exponentials Matter: LTI Systems

The key reason is that complex exponential vectors are eigenvectors of linear time-invariant systems.

The instructor said this is especially familiar to electrical engineers, and suggested that students from other backgrounds may need to connect it to discrete-time signals and systems.

Linearity

A system is linear if it satisfies scaling and superposition.

Scaling:

If input x produces output y , then input αx produces output αy .

Superposition:

If x_1 produces y_1 and x_2 produces y_2 , then

$$\alpha_1 x_1 + \alpha_2 x_2$$

produces

$$\alpha_1 y_1 + \alpha_2 y_2.$$

Shift Invariance / Time Invariance

In the finite-dimensional discrete setting, the instructor described a shift as a circular shift or rotation:

- One entry moves to the top.
- The other entries move down.

If the input is shifted, the output should be shifted in the same way.

That property is time invariance in this finite-dimensional discrete-time context.

Complex Exponentials as Eigenvectors

For a linear time-invariant system, if a complex exponential is given as input, the output is the same complex exponential multiplied by a scalar.

The scalar depends on the frequency.

Geometrically:

- The direction of the vector does not change.
- The vector is scaled by a complex number.

This is exactly eigenvector behavior.

[Exam note] This property is important because it makes the output easy to compute. For each complex exponential component, one only needs to know the scaling factor.

Fourier Transform as Basis Change

If an input can be represented as a linear combination of complex exponential vectors,

$$x = x'_1 f_1 + x'_2 f_2 + \cdots + x'_n f_n,$$

then for an LTI system whose response to f_k is $H_k f_k$, linearity gives

$$\text{output} = x'_1 H_1 f_1 + x'_2 H_2 f_2 + \cdots + x'_n H_n f_n.$$

So the output is easy to compute once the input is represented in the complex exponential basis.

The instructor summarized this as the reason electrical engineers spend significant time learning how to write arbitrary vectors/signals as sums of complex exponentials.

This representation is the **Fourier transform**.

Important interpretation:

The Fourier transform is a basis change.

- Time-domain samples are coordinates with respect to the standard basis.
- Frequency-domain coefficients are coordinates with respect to the complex exponential basis.

The complex exponential basis is useful because LTI systems are easier to analyze in that basis.

The instructor said the course would return to this topic later.

Return to the Main Problem: $Ax = b$

After reviewing span, linear independence, bases, dictionaries, and examples of bases, the instructor returned to the main problem:

$$Ax = b.$$

The goal is to analyze existence and uniqueness of solutions.

[Exam note] Existence and uniqueness mean:

- Given A and b , is there a solution x ?
- If there is a solution, is it unique?
- Or are there multiple solutions?

The instructor emphasized that the analysis uses vector space methods and depends on two different ways of perceiving the matrix-vector product Ax .

Two Views of Matrix-Vector Multiplication

The instructor called these two views simple but fundamental.

They are based on whether A is treated as:

- A collection of column vectors.
- A collection of row vectors.

These two views lead to different analyses:

- Column view: important for existence.
- Row view: important for uniqueness.

First View: A as a Collection of Columns

Let A be an $m \times n$ matrix. Write it as columns:

$$A = [c_1 \ c_2 \ \cdots \ c_n].$$

Each column c_i is an m -dimensional vector.

For

$$x = (x_1, \dots, x_n)^T,$$

the product is

$$Ax = x_1 c_1 + x_2 c_2 + \dots + x_n c_n.$$

[Exam note] Multiplying a matrix on the right by a vector is taking a linear combination of the columns of the matrix.

The instructor justified this through partitioned multiplication: view the matrix as column blocks and the vector entries as scalar blocks. Matrix multiplication then becomes the first column block times x_1 , plus the second column block times x_2 , and so on.

This is the first key perception:

$$Ax = \text{linear combination of columns of } A.$$

The instructor emphasized that this is simple but important.

This view will drive the existence analysis.

Second View: A as a Collection of Rows

The same matrix A can be written in terms of its rows.

The instructor used notation where r_i is conceptually a column vector, so the actual row is r_i^T . Thus rows are written as

$$r_1^T, r_2^T, \dots, r_m^T.$$

This notation can be misleading because r_i is a column vector even though it represents row data after transposition.

Using the row view,

$$Ax = \begin{bmatrix} r_1^T x \\ r_2^T x \\ \vdots \\ r_m^T x \end{bmatrix}.$$

[Exam note] Matrix multiplication can be viewed as taking inner products of the rows of A with the vector x .

Using the Euclidean inner product,

$$r_i^T x = \langle r_i, x \rangle.$$

So the row view says:

$$Ax = \text{the vector of inner products of } x \text{ with the rows of } A.$$

This view will be important for uniqueness.

Left Multiplication by a Vector

The instructor also discussed the dual situation: multiplying A from the left by a row vector y^T :

$$y^T A.$$

For dimensions to match, if A is $m \times n$, then y has length m and $y^T A$ is a row vector of length n .

If A is viewed as a collection of row vectors, then $y^T A$ is a linear combination of the rows of A .

So:

- Right multiplication Ax gives linear combinations of columns.
- Left multiplication $y^T A$ gives linear combinations of rows.

If A is viewed as a collection of columns, then $y^T A$ gives inner products of y with the columns of A :

$$y^T A = [y^T c_1 \quad y^T c_2 \quad \cdots \quad y^T c_n].$$

The instructor briefly became tangled in notation but returned to the main summary:

- Right multiply: linear combination of columns, or inner products with rows.
- Left multiply: linear combination of rows, or inner products with columns.

Why the Two Views Matter

[Exam note] The instructor gave the punchline:

The column and row perceptions are important because the course will define subspaces based on them.

These subspaces will then be used to analyze existence and uniqueness.

Specifically:

- Existence will be addressed using the column/range space.
- Uniqueness will later be addressed using row-related or nullspace-related ideas.

When a student asked how these views explain existence and uniqueness, the instructor said this section was preparation: the course first builds the relevant subspaces, then uses them for the two questions.

The current lecture begins the existence side.

Geometric Picture of $Ax = b$

The instructor described Ax as a function or mapping.

For an $m \times n$ matrix A :

$$A : \mathbb{R}^n \rightarrow \mathbb{R}^m.$$

The input x is n -dimensional.

The output Ax is m -dimensional.

The right-hand side b also lies in the m -dimensional output space.

Instructor warning:

The symbols m and n are easy to confuse, but the convention here is:

- n : input dimension and number of columns.
- m : output dimension and number of rows.

The instructor noted that this m, n convention is standard in many references.

Instructor correction:

The instructor initially almost swapped the roles of m and n , then corrected the picture: x is n -dimensional and Ax , like b , is m -dimensional. The drawings are only toy geometric pictures, not necessarily actual three-dimensional-to-three-dimensional maps.

$Ax = b$ as an Inverse Problem

Solving

$$Ax = b$$

is an inverse problem.

The output b is known. The goal is to find an input x that generated it.

Question:

Which x , if any, maps to the given b ?

This interpretation is central to the existence question.

If no input maps to b , then the equation has no solution.

So an $Ax = b$ picture should be read as a search for a preimage of b , not as a guarantee that the displayed target is actually attainable.

Where Rows and Columns Live

For an $m \times n$ matrix A :

- Each row has length n , so rows live in the input space $\mathbb{R}^n$.
- Each column has length m , so columns live in the output space $\mathbb{R}^m$.
- The vector x lives in the input space $\mathbb{R}^n$.
- The vector b lives in the output space $\mathbb{R}^m$.

[Exam note] This matches the two product views:

- Rows must be in the same space as x , because Ax computes inner products with rows.
- Columns must be in the same space as b , because Ax is a linear combination of columns and therefore produces an output-space vector.

This is the fundamental geometric picture for analyzing $Ax = b$.

Column Space / Range Space

The first vector space introduced for existence analysis is the **column space**.

Definition:

The column space of A is the span of the columns of A .

$$\text{Col}(A) = \text{span}\{c_1, \dots, c_n\}.$$

Because the columns are m -dimensional, the column space is a subspace of the output space:

$$\text{Col}(A) \subseteq \mathbb{R}^m.$$

The instructor also called this the **range space**.

Geometric picture:

The instructor drew the column space as a subspace, such as a plane inside the output space. He cautioned that the drawing was only a toy picture; if the drawn columns look linearly dependent, that visual dependence should not be over-interpreted.

Warning about notation:

The transcript says the textbook/instructor may use $R(A)$ for range space. This can be confused with row space, but here $R(A)$ means range/column space.

Column Space as All Possible Outputs

Using the column view,

$$Ax = x_1c_1 + \cdots + x_nc_n.$$

As x varies over all possible inputs, Ax varies over all possible linear combinations of the columns.

Therefore:

$$\text{Col}(A) = \{Ax : x \in \mathbb{R}^n\}.$$

This is the image of the input space inside the output space.

Interpretation:

The column/range space is the set of all outputs that the linear mapping $x \mapsto Ax$ can generate.

Existence Criterion for $Ax = b$

The first major conclusion:

$$Ax = b \text{ has at least one solution} \iff b \in \text{Col}(A).$$

The instructor stated it in two parts:

- If b lies in the column/range space of A , then there is at least one x such that $Ax = b$.
- If b does not lie in the column/range space, then there is no solution.

Reason:

The only vectors A can produce are linear combinations of its columns. If b is not one of those vectors, no choice of x can produce it.

Procedure for checking existence conceptually:

1. Take the columns of A .
2. Find or understand their span.
3. Check whether b belongs to that span.

The instructor emphasized that this analysis comes directly from viewing A as a collection of columns.

When Every b Has a Solution

The next question:

For which matrices A does $Ax = b$ have a solution for every possible $b \in \mathbb{R}^m$?

This is equivalent to asking when the column/range space fills the entire output space:

$$\text{Col}(A) = \mathbb{R}^m.$$

If this holds, then every $b \in \mathbb{R}^m$ lies in the range space, so every right-hand side has at least one solution.

Important limitation:

This only says something about existence. It does not imply uniqueness.

Instructor remark:

The lecture connects this condition to full rank. The instructor noted that although this looks like a column-space property, it will be related to independence of rows, specifically full row rank.

[Likely exam topic] This is the bridge to the later full-row-rank condition: for an $m \times n$ matrix to have $\mathcal{R}(A) = \mathbb{R}^m$, the rows will have to supply m independent constraints. The instructor explicitly warned that “independent columns” is not the right automatic answer for this existence-for-every- b property.

Matrix Shapes and Relative Dimensions

[Exam note] The relative dimensions of the matrix are important.

For an $m \times n$ matrix:

- $m > n$: tall matrix, more rows than columns.
- $m = n$: square matrix.
- $n > m$: fat/wide matrix, more columns than rows.

The instructor used the terms tall and fat. The transcript also mentions that some references may use “skinny” for tall matrices.

Necessary Condition for Full Range

To have

$$\text{Col}(A) = \mathbb{R}^m,$$

the columns of A must span an m -dimensional output space.

Since A has only n columns, a necessary condition is:

$$n \geq m.$$

Interpretation:

There must be at least as many columns as the dimension of the output space.

The instructor gave the example:

If $m = 100$ and $n = 10$, then we have ten vectors in a 100-dimensional space. Ten vectors cannot span all of $\mathbb{R}^{100}$.

Therefore, a tall matrix $m > n$ cannot have column space equal to all of $\mathbb{R}^m$. There will be some b 's outside the range space, so one cannot guarantee a solution for arbitrary b .

Important warning:

$$\text{Col}(A) = \mathbb{R}^m \Rightarrow n \geq m.$$

This is a one-way implication. It is necessary but not sufficient.

Not every square or fat matrix has full range. Having enough columns does not guarantee that the columns actually span the entire output space.

Input and Output Dimension Interpretation

The condition $n \geq m$ means:

The input space must have dimension at least as large as the output space if we want the possibility of generating every output.

Again, this is only necessary. Additional rank/independence conditions are needed.

Alternative Definitions of Range Space

The instructor summarized three equivalent descriptions of the range space of A .

1. Image Definition

$$\mathcal{R}(A) = \{Ax : x \in \mathbb{R}^n\}.$$

This is the set of all outputs generated by all possible inputs.

2. Column Space Definition

$$\mathcal{R}(A) = \text{span}\{c_1, \dots, c_n\}.$$

This is the span of the columns of A .

3. Solvability Definition

$$\mathcal{R}(A) = \{b \in \mathbb{R}^m : Ax = b \text{ has at least one solution}\}.$$

This is the collection of right-hand sides for which the linear system is solvable.

These definitions are the same because Ax is exactly a linear combination of the columns of A .

Full Range and Right Inverses

[Likely exam topic] The instructor then introduced a two-way implication involving full range and right inverses.

The condition

$$\mathcal{R}(A) = \mathbb{R}^m$$

is equivalent to the existence of a matrix $D \in \mathbb{R}^{n \times m}$ such that

$$AD = I_m.$$

The transcript uses both Z and D language, but the constructed matrix is effectively D .

The dimensions are the transpose shape of A : if A is $m \times n$, then a right inverse D must be $n \times m$. Thus for a fat A , the right inverse D is tall.

Such a matrix D is called a **right inverse** of A , because it multiplies A on the right and produces the identity:

$$AD = I_m.$$

Important warning:

This does not claim that

$$DA = I_n.$$

For non-square matrices, right and left inverse properties are different.

Proof Idea: Full Range Implies a Right Inverse

Assume

$$\mathcal{R}(A) = \mathbb{R}^m.$$

Then every $b \in \mathbb{R}^m$ has at least one solution to $Ax = b$.

Now choose the standard basis vectors of the output space:

$$e_1, e_2, \dots, e_m.$$

Since each $e_i \in \mathbb{R}^m$, and the range is all of $\mathbb{R}^m$, each system

$$Ax = e_i$$

has at least one solution.

Pick one solution for each standard basis vector:

$$Ad_1 = e_1, \quad Ad_2 = e_2, \quad \dots, \quad Ad_m = e_m.$$

If there are multiple choices, choose any one; existence is what matters here.

Now assemble these solution vectors as columns:

$$D = [d_1 \ d_2 \ \cdots \ d_m].$$

Then

$$AD = A[d_1 \ d_2 \ \cdots \ d_m] = [Ad_1 \ Ad_2 \ \cdots \ Ad_m] = [e_1 \ e_2 \ \cdots \ e_m] = I_m.$$

So D is a right inverse of A .

This proof uses the column-partition view of matrix multiplication again: multiplying A by a matrix D column by column gives the columns Ad_i .

Converse Proof Idea: Right Inverse Implies Full Range

The instructor announced the implication as two-way. The immediate converse is:

If there exists D such that

$$AD = I_m,$$

then for any $b \in \mathbb{R}^m$, choose

$$x = Db.$$

Then

$$Ax = A(Db) = (AD)b = I_m b = b.$$

So every b has a solution, which means

$$\mathcal{R}(A) = \mathbb{R}^m.$$

This completes the equivalence between full range and existence of a right inverse.

Right Inverse, Square Matrices, and Non-Square Matrices

The instructor made several important remarks about right inverses.

Square Case

If A is square and a right inverse D exists, then:

- The right inverse is unique.
- It will also turn out to be a left inverse.
- So $DA = I$ as well.
- This unique matrix is the usual inverse A^{-1} .

The instructor said this would be shown later.

Fat Matrix Case

If A is fat/wide and has full range, then right inverses can exist.

In that case, the right inverse is not unique. There can be infinitely many such matrices D satisfying

$$AD = I_m.$$

This aligns with the idea that fat systems often have more input degrees of freedom than output constraints.

Tall Matrix Case

A tall matrix cannot have full range when $m > n$, because it does not have enough columns to span $\mathbb{R}^m$.

Therefore, tall matrices cannot have a right inverse of this type.

General Non-Square Case

For non-square matrices, one talks about right inverses and left inverses separately.

The usual unique inverse concept belongs to the square case.

Relationship Between Concepts

The lecture repeatedly ties advanced-looking matrix ideas back to simple vector space ideas.

Key relationships:

- Span is the set of all linear combinations.
- A basis is a spanning set without redundancy.
- Linear independence formalizes “no redundancy.”
- A dictionary is a spanning set that may contain redundancy.
- Redundancy produces non-unique coefficients.
- Non-unique coefficients can be used for sparsity.
- Coordinates are coefficients relative to a chosen basis.
- Changing basis changes coordinates but not the vector itself.
- Fourier transform is a basis change from standard/time coordinates to complex exponential/frequency coordinates.
- Ax as a column combination connects matrix multiplication to span.
- The column space/range space is the set of all possible outputs of $x \mapsto Ax$.
- $Ax = b$ is solvable exactly when b lies in that range.
- Every b is solvable exactly when the range is all of $\mathbb{R}^m$.
- Full range requires $n \geq m$, but this is not sufficient.
- Full range is equivalent to existence of a right inverse D with $AD = I_m$.

Instructor Warnings and Hints

[Likely exam topic] Existence and uniqueness questions are important, and they motivate the vector space machinery.

[Exam note] The column-space/range-space view is central to existence.

[Exam note] The row-space/inner-product view will be central to uniqueness later.

[Exam note] Remember both matrix-vector product perceptions:

- Ax as a linear combination of columns.
- Ax as inner products with rows.

[Exam note] Relative dimensions m and n matter. Be careful:

- m : number of rows and output dimension.
- n : number of columns and input dimension.

Warning:

The geometric sketches are intuition only. Always check the actual dimensions: rows live with x in the input space, columns live with b in the output space.

Warning:

The condition $n \geq m$ is necessary for full range, but it is not sufficient.

Warning:

Full range is not obtained merely by saying “the columns are independent.” The lecture flags full row rank as the relevant rank language for guaranteeing every $b \in \mathbb{R}^m$ is reachable.

Warning:

Do not confuse range-space notation $R(A)$ with row space. In this lecture, $R(A)$ refers to range/column space.

Warning:

A right inverse satisfies $AD = I_m$, not necessarily $DA = I_n$.

Warning:

Orthogonal matrices form a subset of square matrices but not a subspace, because the zero matrix is not included.

Instructor remark:

The complex exponential basis will be examined in homework and revisited later. Its importance comes from LTI systems and Fourier analysis.

Instructor closing remark:

The concepts of right inverse and solvability are built from the basic concepts of linear combination, span, and vector spaces. The instructor asked students to review this before the next lecture.

Source / Coverage Note

These notes were created only from `corrected/lecture5_corrected.md`. They cover the chronological lecture content, including definitions, examples, proof ideas, instructor remarks, warnings, and the transcript's exam-relevant markers through the closing review instruction. No other lecture was processed.